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Chapter 34

GELFAND-P001520

15/$\overline{V}$-89. Gelfand Sem.

V. Drinfeld.

Quasi-Hopf algebras and the Knizhnik–Zamolodchikov equations.

The Kn.–Zam. eqs. arose in conformal field theory

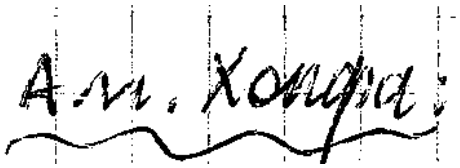
Theorem [above the line, pointing to “Theorem”: (Kohno and somebody else)] on the equiv. of two classes of reprs. of the Artin braid group.

1st class — matrices

2nd class — monodromy of the Kn.–Zam. eqs.

Hopf alg.: an assoc. algebra A with an algebra homomorphism

$$\Delta\colon A \longrightarrow A \otimes A$$



Uncertain reading. alg.

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there are axioms for the counit and antipode

Coassociativity: the diagram commutes

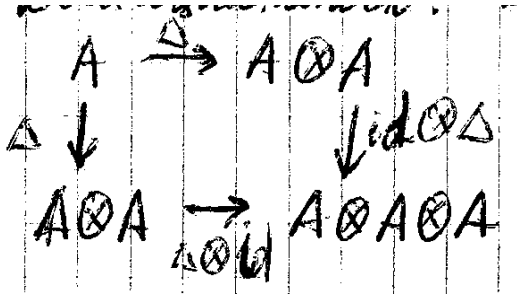


Diagram and typed description.

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \otimes A \\ {\scriptstyle\Delta}\downarrow & & \downarrow{\scriptstyle id \otimes \Delta} \\ A \otimes A & \xrightarrow[\Delta \otimes id]{} & A \otimes A \otimes A \end{array}$$

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or

$$(id \otimes \Delta)(\Delta(a)) = (\Delta \otimes id)(\Delta(a)) \tag{1}$$

Quasi-Hopf property: equality (1) holds up to conjugation

$$(id \otimes \Delta)(\Delta(a)) = \phi \cdot (\Delta \otimes id)(\Delta(a))\phi^{-1}, \tag{2}$$

where $\phi \in A \otimes A \otimes A$

5-gon axiom

$$\begin{aligned} &(id \otimes id \otimes \Delta)(\phi) \cdot (\Delta \otimes id \otimes id)(\phi) \\ &\quad = (1 \otimes \phi) \cdot (id \otimes \Delta \otimes id)(\phi)(\phi \otimes 1) \end{aligned} \tag{3}$$

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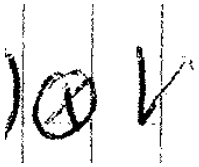
[source number: 36]

Meaning of axiom (3): the composition

$$A \xrightarrow{\Delta} A \otimes A \longrightarrow \mathrm{End}(V_1 \otimes V_2) \tag{4}$$

will be a representation (if the V_j are representations), and there is weak associativity: an isomorphism

$$(V_1 \otimes V_2) \otimes V \xrightarrow{\sim} V_1 \otimes (V_2 \otimes V_3) \tag{5}$$

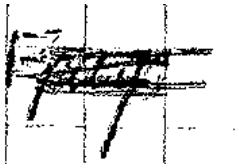


Uncertain reading. V

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[Below the arrow, pointing to it:] not the identity! (constructed using ϕ).

Suppose there are 4 factors. Then there are 5 ways to place the parentheses.



Undeciphered passage. Crossed out before the diagram: a short word, the letters obscured by strokes.

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We draw the diagram:

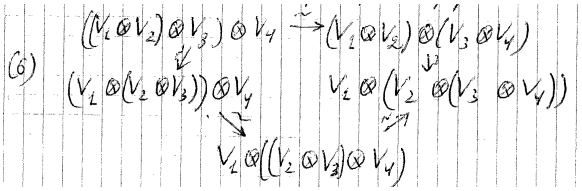


Diagram and typed description.

$$\begin{array}{ccc} ((V_1 \otimes V_2) \otimes V_3) \otimes V_4 & \xrightarrow{\sim} & (V_1 \otimes V_2) \otimes (V_3 \otimes V_4) \\ \downarrow \sim & & \downarrow \sim \\ (6)\ (V_1 \otimes (V_2 \otimes V_3)) \otimes V_4 & & V_1 \otimes (V_2 \otimes (V_3 \otimes V_4)) \\ & \searrow^{\sim} \quad \swarrow_{\sim} & \\ & V_1 \otimes ((V_2 \otimes V_3) \otimes V_4) & \end{array}$$

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(all the isomorphisms are constructed using ϕ).

Requirement: diagram (6) commutes.

Quasitriangular quasi-Hopf algebras.

A / Δ [sign heavily overwritten] is commutative: $R\Delta = \Delta R$, $R \in A \otimes A$



Uncertain reading. A / Δ

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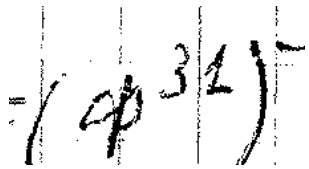
Uncertain reading. $R\Delta = \Delta R$

One requires $\Delta'(a) = R\Delta(a)R^{-1}$ — also a comultiplication

6-gon axioms

$$(\Delta \otimes id)(R) = \Phi^{312}R^{13}(\Phi^{132})^{-1}R^{23}\phi \tag{7}_1$$

$$(id \otimes \Delta)(R) = (\phi^{31})^{-1}R^{13}\phi^{213}R^{12}\Phi^{-1} \tag{7}_2$$



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Uncertain reading. 31

In fact quasi-Hopf algebras are simpler [inserted above the line: (than Hopf algebras)], since there is an operation of Twisting. [Below "Twisting": ($\sim$ gauge transformation)] Let there be given A, Δ, ϕ (a quasi-Hopf algebra)

Suppose there is an invertible element $F \in A \otimes A$

[I.M.G.'s question: a) what is the dual object? — dual space



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Uncertain reading. dual object

b) what is the quasi-Hopf property on the dual space?]

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The new quasi-Hopf algebra:

$$\widetilde{\Delta}(a) = F \cdot \Delta(a) \cdot F^{-1}, \tag{8}$$

$$\widetilde{\phi} = F^{23}(id \otimes \Delta)(F) \cdot \phi(\Delta \otimes id)(F^{-1})(F^{12})^{-1}, \tag{9}$$

$$\widetilde{R} = F^{21}RF^{-1}. \tag{10}$$

It is natural to regard such objects as equivalent.



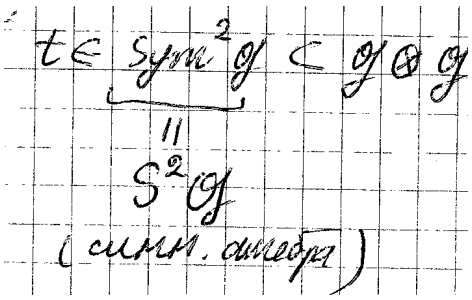
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Next, to study Hopf algebras it is useful to consider quasi-Hopf algebras.

Construction of quasi-Hopf algebras.

Initial data: Lie alg. g,



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Diagram and typed description.

$t \in \underbrace{\mathrm{Sym}^2 g}_{\|\ S^2 g} \subset g \otimes g$

(symm. algebra)

$$A = (U_g)[[h]] \quad \text{— an algebra over } \mathbb{C}[[h]] \tag{11}$$

[Below U_g, with a connecting stroke: “univ. enveloping”.]

Comultiplication

$$x \in g \Rightarrow \Delta(x) = x \otimes 1 + 1 \otimes x \tag{12}$$

$$R = e^{ht/2} \tag{13}$$

and now to determine ϕ one writes down a tricky system of eqs. which has a solution, unique up to twists that leave the form of the comultiplication and t unchanged.

Consider the ordinary diff. eq.:

$$G'(x) = \bar{h}\left(\frac{t^{12}}{x} + \frac{t^{23}}{x-1}\right) G(x), \quad \bar{h} = \frac{h}{2\pi i}, \tag{14}$$

where

$$G(x) \in (U_g \otimes U_g \otimes U_g)[[h]].$$

This construction yields all deformations of quasi-Hopf algebras

Now ϕ is determined as follows: on $(0,1)$ consider 2 solutions

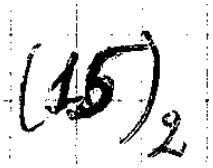
GELFAND-P001523

[source number: 37]

G_j, $j = 1, 2$, with standard asymptotics at 0 and 1:

$$G_1(x) \sim x^{\bar{h}t^{12}}, \quad x \to 0 \tag{15}_1$$

$$G_2(x) \sim (1-x)^{\bar{h}t^{23}}, \quad x \to 1 \tag{15}_2$$



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Uncertain reading. Number of the second line: the digit 5 is retraced or corrected.

Then

$$G_1 = G_2 \cdot \Phi, \quad \phi \in (U_g \otimes U_g \otimes U_g)[[h]]. \tag{16}$$

and this ϕ satisfies all the required conditions.

I.M.G. The Kn.–Zam. eqs. are a very simple and deep example of a hypergeom. function.

Dotsenko. Ising model.

Motivated by an old work of A.M. Polyakov on the 3-dimensional Ising model

The Ising model on a lattice of any dimension: the partition function has the form

$$Z(\beta) = \sum_{\{\sigma\}} \exp\left\{-\beta \sum_{x,\alpha} \sigma_x \sigma_{x+\alpha}\right\} = \sum_{\{\sigma\}} \prod_{x,\alpha} e^{-\beta \sigma_x \sigma_{x+\alpha}} \tag{1}$$

$$x \in \mathbb{Z}^d; \qquad \alpha \in \mathbb{Z}^d, \quad |\alpha| = 1, \qquad \beta > 0$$

$\{\sigma\}$ — spin configurations, $\sigma_x = \pm 1$.

(Partition sum = Partition function)

Hamiltonian:

[K. Khanin]

$$H = -\sum_{x,\alpha} \sigma_x \sigma_{x+\alpha} \tag{2}$$

One should take everything in a finite container V and the free energy

$$f(\beta) = \lim_{V \to \infty} \frac{\ln Z_V(\beta)}{|V|} \tag{3}$$

is independent of the boundary conditions and determines phase transitions

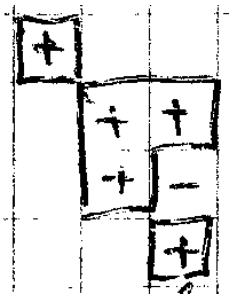
One can also take

$$H = -\sum_{x,\alpha} \sigma_x \sigma_{x+\alpha} + h \sum_x \sigma_x \tag{4}$$

and calculate correlation functions.

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A configuration [inserted above the line: (for $d = 2$)] is uniquely characterized by a set of non-intersecting contours (what separates the pluses and minuses).



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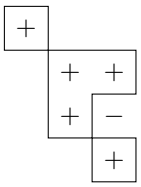


Diagram and typed description.

It is easy to check that in the 2-dimensional case

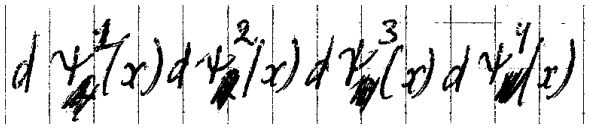
$$Z(\beta) = \sum_{\text{contours}} (\lambda)^L; \tag{5}$$

where

$$\lambda = e^{-2\beta}; \qquad L \text{ — contour length}$$

Next, one can write (from (5))

$$Z(\beta) = \prod_x \int d\psi^1(x)\, d\psi^2(x)\, d\psi^3(x)\, d\psi^4(x)\ \exp\{A[\psi]\}, \tag{6}$$



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Uncertain reading. Crossed-out subscripts under the four ψ symbols: reading undetermined.

where ψ^α — Grassmann variables (4 at each site);



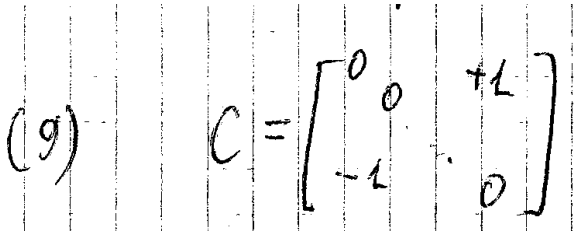
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$$A[\psi] = -\frac{1}{2}\sum_{x,\alpha}(\psi(x)\, C\, \psi(x)) + \lambda \sum_{x,\alpha} \psi(x+\alpha) P_\alpha \psi(x) \tag{7}$$

$$(\psi C \psi) = \psi^\alpha C^{\alpha\beta} \psi^\beta; \tag{8}$$

$C^{\alpha\beta}$ — a certain matrix (not uniquely determined)



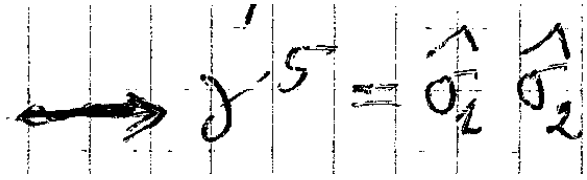
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Diagram and typed description.

(9) $$C = \begin{bmatrix} 0 & 0 & +1 \\ & 0 & \\ -1 & & 0 \end{bmatrix}$$

In the continuum limit

$$C \longleftrightarrow \gamma^5 = \hat{\sigma}_1 \hat{\sigma}_2 \tag{10}$$



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Uncertain reading. γ^5

σ_j — Pauli matrices

$$P_\alpha = \frac{1 + \vec{\alpha}\,\vec{\sigma}}{2} \quad \text{— projector} \tag{11}$$

The contours reduce to directed walks. In the walk a factor $\lambda e^{i\Delta\varphi/2}$ arises, which

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[source number: 38]

enters the amplitude (complex probability); $\Delta\varphi$ — angle of turn.

$e^{i\Delta\varphi/2}$ — Kac–Ward factors

Explanation by a walk problem — Feynman.

$$\sum_{\mathcal{P}_{x_1,x_2}} \prod \lambda e^{i\Delta\varphi/2} = \langle \psi^\alpha(x_1)\psi^\beta(x_2)\rangle \tag{12}$$



Diagram and typed description. $= \sum_{\mathcal{P}_{x_1,x_2}}$



$=$

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$$= W^{\alpha\beta}(x_1, x_2)$$

$$\mathcal{P}_{x_1,x_2} = \{\text{paths from } x_1 \text{ to } x_2\}$$

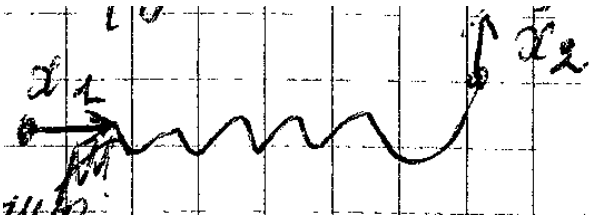


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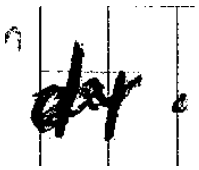


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By definition:

$$\int d\psi = 0, \qquad \int d\psi \cdot \psi = 1 \tag{13}$$



Uncertain reading. $d\psi$

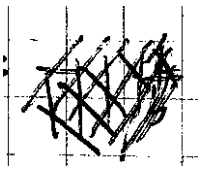
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Parafermionic variables are sometimes needed (see V. Fateev, A. Zamolodchikov, ZhETF [below the journal title: 1985]). [Below the line:] constructed explicitly in conformal field theory

Remark. Calculation of the specific heat in repr. (6)

Set

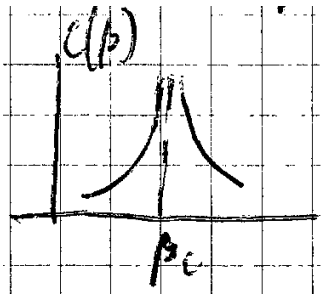
$$F(\beta) = \ln Z(\beta) \tag{14}$$



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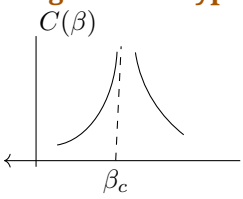
Uncertain reading. Crossed out before $\ln Z(\beta)$: reading undetermined.

$$\frac{d^2F(\beta)}{d\beta^2} \sim \frac{Z''}{Z} - \frac{(Z')^2}{Z^2} = \underset{\text{specific heat}}{\underline{\underline{C(\beta)}}} \sim (\beta - \beta_c)^{-\alpha} \tag{15}$$



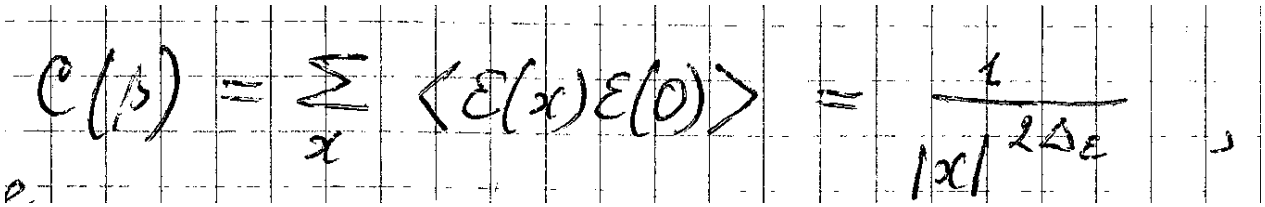
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Diagram and typed description.



The exponent α is a critical index. We have:

$$C(\beta) = \sum_x \langle \varepsilon(x)\varepsilon(0)\rangle = \frac{1}{|x|^{2\Delta_\varepsilon}}, \tag{16}$$

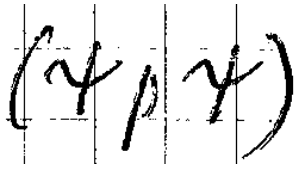


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Uncertain reading. $\frac{1}{|x|^{2\Delta_\varepsilon}}$

where

$$\varepsilon = (\psi\,\rho\,\psi)$$



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Uncertain reading. ρ

Here Δ_ε is uniquely related to α and is called the anomalous dimension:

$$\alpha = \frac{d - 2\Delta_\varepsilon}{d - \Delta_\varepsilon}, \qquad d \text{ — dimension of space} \tag{17}$$

(the usual scaling relation)

3-dimensional case.

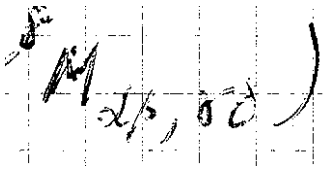
$$Z(\beta) = \prod_{\text{edges}} \int d\psi^\alpha_{x,\mu} \, \exp\{A[\psi]\} \tag{18}$$

$$\begin{aligned} A[\psi] = {} & \frac{1}{2} \sum_{\text{edges}} \left(\psi^\alpha C^{\alpha\beta} \psi^\beta \right) \\ & + \lambda \sum_{\text{faces}} \left(\psi^\alpha \psi^\beta \psi^\gamma \psi^\delta M_{\alpha\beta,\gamma\delta \,/\, \delta\gamma} \right) \end{aligned} \tag{19}$$



Uncertain reading. β

Original source details: GELFAND-P001526



Uncertain reading. $\gamma\delta \,/\, \delta\gamma$

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[In the first term the index of the last ψ and the closing parenthesis are heavily overwritten.]

$$Z(\beta) = \sum_S \lambda^{A(S)}, \tag{20}$$

where S — closed surface in R^3,

(encl. different phases)



Uncertain reading. encl.

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$\psi^\alpha_{x,\mu}$ — Grassmann variables

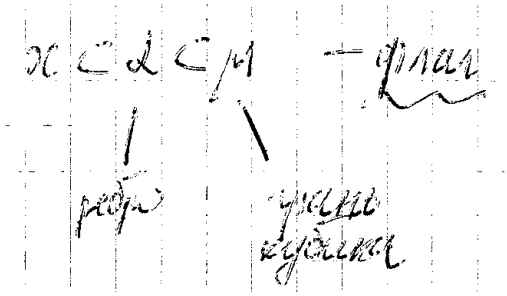


Diagram and typed description.

$x \subset \alpha \subset \mu$ — flag

edge

face of a cube

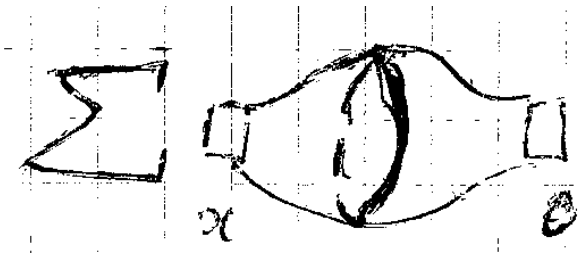
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Let us calculate

$$\ln Z = F, \tag{21}$$

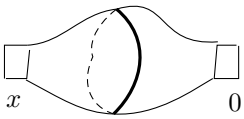
then

$$F'' = \sum \langle \varepsilon(x)\varepsilon(0)\rangle \tag{22}$$



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Diagram and typed description. $= \sum$



[The thick strokes at the junctions are indistinct.]

(birth of a closed fermionic string)

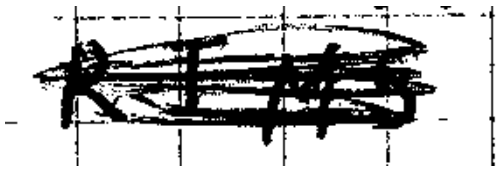
GELFAND-P001527

[source number: 39]

Next one has to solve the problem in the continuum limit.

References: Dotsenko, Polyakov (~ 60 pp.)

(in the book Advanced Studies in Pure Mathematics,



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Undeciphered passage. Crossed out below "in the book": a short fragment, the letters obscured by strokes.

v. 16, 1988) [The first digit of the year is heavily retraced.]



Original source details: GELFAND-P001527

Uncertain reading. Crossed-out sign between Pure and Mathematics: &